

ALENT SIBY

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PROFILE

Electronics and Computer Science Engineering student with a strong foundation in embedded systems, IoT, and full-stack development, currently pursuing a minor in Robotics and Automation. Experienced in building hardware-software integrated solutions, including safety systems, assistive devices, and web-controlled automation platforms. Strong analytical thinker with a practical, solution-oriented approach.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript, Java, HTML, CSS, SQL

Hardware & Embedded: ESP32/ESP8266, Arduino, Raspberry Pi, PCB design, circuit simulation

Communication & Protocols: I2C, SPI, MQTT

Tools & Modeling: Firebase, Git, MATLAB, Fusion 360 (CAD), Selenium, Proteus, EasyEDA, Arduino IDE, LaTeX, Canva

Frameworks & Libraries: Flutter, React, Electron, OpenCV, MediaPipe, Node.js, Streamlit

Soft Skills: Leadership, time management, adaptability, continuous learning

Languages: Malayalam, English

EDUCATION

B.Tech in Electronics and Computer Science Engineering 2023 - 2027
St. Joseph's College of Engineering and Technology, Palai (Autonomous), Kottayam
CGPA: 7.0

Higher Secondary Education (12th) 2022 - 2023
Marygiri CMI Public School, Koothattukulam
Percentage: 78%

Secondary Education (10th) 2020 - 2021
Vijayamatha Public School, Thookkupalam, Idukki, Kerala
Percentage: 90%

EXPERIENCE

Full-Stack Web Development Intern | Main Flow Services Jun 2025 - Aug 2025

- Developed a home automation platform using PHP and MySQL for backend architecture.
- Implemented Firebase-based data handling to enable real-time dashboard features and remote device control.
- Designed secure user interaction flows, including role-based authentication and basic system testing.

PROJECTS

IoT Smart Gas Stove Retrofit System

- Built an ESP32-based safety system for gas stoves with gas leak detection and whistle detection logic.
- Integrated temperature sensing and a servo mechanism to control the gas valve using a regulated power supply.
- Enabled remote monitoring through Blynk with manual override control.

Eye-Controlled Environment System (For Paralyzed Users)

- Implemented an eye-tracking system using Python with OpenCV and MediaPipe to detect gaze and trigger control actions.

Refreshable Braille Display

- Developed an assistive communication prototype using ESP32, solenoids, and shift registers to translate digital text into tactile Braille patterns.
- Built a companion Flutter Bluetooth app to send real-time text to the hardware display.

Live Production Software (Desktop Application)

- Built a desktop application inspired by live production tools using Electron and React.
- Implemented media input handling, canvas-based rendering, audio mixing, and recording functionality.

Sim Racing Rig

- Designed and assembled the mechanical framework and control interfaces for a custom racing simulator.
- Programmed an Arduino-based control system to capture steering, paddle, and pedal input.

Analog Electronics Projects

- Built a transistor-based VU meter and an LM386 audio amplifier to strengthen understanding of analog signal processing.

CERTIFICATIONS

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| ● NPTEL Online Certification on Fundamentals of Artificial Intelligence,
IIT Guwahati (12 weeks) | Jul 2025 - Oct 2025 |
| ● NPTEL Online Certification on Wheeled Mobile Robots, IIT Madras and
IIT Palakkad (8 weeks) | Jan 2026 - Mar 2026 |
| ● Drone Assembly & Flight Workshop ISTE SJCET | Mar 2025 |